THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 5: **Exploring Forces** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Ball bearings are used to: 1

(a) increase sliding friction **(b)** increase weight
(c) stop motion **(d)** change sliding into rolling friction

2. A charged body can be detected by a: 1

(a) spring balance **(b)** stopwatch
(c) gold-leaf electroscope **(d)** thermometer

3. Which of the following is a non-contact force? 1

(a) gravitational force **(b)** force of a push by hand
(c) friction **(d)** muscular force

4. Friction produces: 1

(a) cold **(b)** heat **(c)** sound only **(d)** light only

5. In an electroscope the leaves open out when: 1

(a) a magnet touches it **(b)** a charged body touches the metal disc
(c) an uncharged body touches it **(d)** water touches it

6. A rubbed plastic comb picks up tiny bits of paper due to: 1

(a) nuclear force **(b)** electrostatic force
(c) muscular force **(d)** friction

7. Static friction acts when: 1

(a) an object is falling **(b)** an object is sliding
(c) an object is at rest but tends to move **(d)** an object is rolling

8. Lightning is caused by: 1

(a) discharge of electric charges between clouds or with the earth
(b) cooling
(c) wind only
(d) sound waves

9. The force that pulls objects towards the centre of the Earth is: 1

(a) electrostatic force **(b)** gravitational force
(c) magnetic force **(d)** friction

10. The friction between a sliding object and the surface is called: 1

(a) air friction **(b)** rolling friction **(c)** sliding friction **(d)** static friction

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do vehicles have treads on the tyres? 2

12. Why does a ball thrown up return to the ground? 2

13. State two effects of force on an object. 2

14. What are balanced forces? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. State three methods of reducing friction. 3

16. Describe the damage caused by lightning and three safety measures. 3

17. Explain why walking on a wet or oily floor is difficult. 3

OR

State three methods of increasing friction. 3

18. What is gravitational force? Give three examples. 3

OR

Differentiate between the force of friction and the gravitational force on the basis of direction and contact. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain with an activity how friction depends on the nature of the surfaces and on the load. 5

OR

Describe how force can change the state of motion and the shape of an object. Include the idea of balanced and unbalanced forces. 5

20. Describe how friction is helpful and harmful with three examples each and how it can be controlled. 5

OR

Discuss rolling, sliding and static friction in terms of their sizes and give daily-life uses of rolling. 5

END OF QUESTION PAPER — Total: 40 Marks

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